

Diagram illustrating a beam of length l and height h . The beam is inclined at an angle θ to the horizontal. A spring with stiffness k is attached to the beam at a distance $\frac{2}{2}$ from the left end. The beam is supported by a vertical reaction force P at the right end. The vertical displacement of the beam at the right end is denoted by δh . The horizontal displacement of the beam at the right end is denoted by δe .

$$e = \frac{r}{2 \cos \theta} - \frac{r}{2}$$

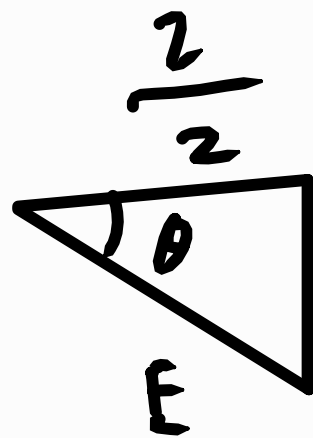
$$= 2 \cos \theta \, \delta \theta$$

$$\delta W = \delta U$$

$$p \delta h = F \delta e$$

$$p(x \cos \theta \cancel{d\theta}) = F \left(\frac{x}{2} \sec \theta + \tan \theta \cancel{d\theta} \right)$$

$$P = \frac{1}{2} F \sec^2 \theta + \tan \theta$$



$$E \cos \theta = \frac{2}{2}$$

$$E = \frac{2}{2 \cos \theta}$$

$$\delta e = \frac{de}{d\theta} \delta \theta$$

$$= \frac{7}{2} \sec \theta \tan \theta \delta \theta$$

$$1) P = \frac{1}{2} F \sec^2 \theta \tan \theta$$

$$P = \frac{1}{2} k \sec^2 \theta \tan \theta$$

$$= \frac{1}{2} k \sec^2 \theta \tan \theta \left(\frac{1}{2} \sec \theta - \frac{1}{2} \right)$$

$$= \frac{1}{4} k \sec^3 \theta \sin \theta (\sec \theta - 1)$$

Diagram illustrating a mechanical system with a mass m on a horizontal surface. The system is connected to a spring and a pulley system. The diagram shows a horizontal line representing a surface. A mass m is on the left. A spring is attached to the mass and a vertical line. A pulley system is on the right, with a mass m hanging from it. The diagram is labeled with various parameters: b , θ , a , z , y , x , δy , δx , m , and ω^2 .

$$\frac{F}{2} = a \sin \theta$$

$$F = 2a \sin \theta$$

$$\delta x = -r \sin \theta \delta \theta \quad \delta y = -r \cos \theta \delta \theta$$

$$\delta y = 2 \cos \theta \delta \theta$$

$$\delta W = \delta U$$

$$2mg\delta y + 2m\omega^2\delta x = F\delta e$$

$$\cancel{2} mg(2 \cos \theta \cancel{\delta \theta}) + \cancel{2} m(b + 2 \cos \theta) \omega^2 (-2 \sin \theta \cancel{\delta \theta}) = F (-\cancel{2} a \cos \theta \cancel{\delta \theta})$$

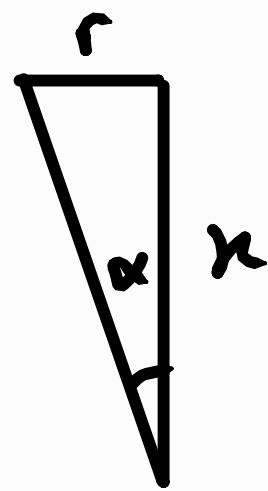
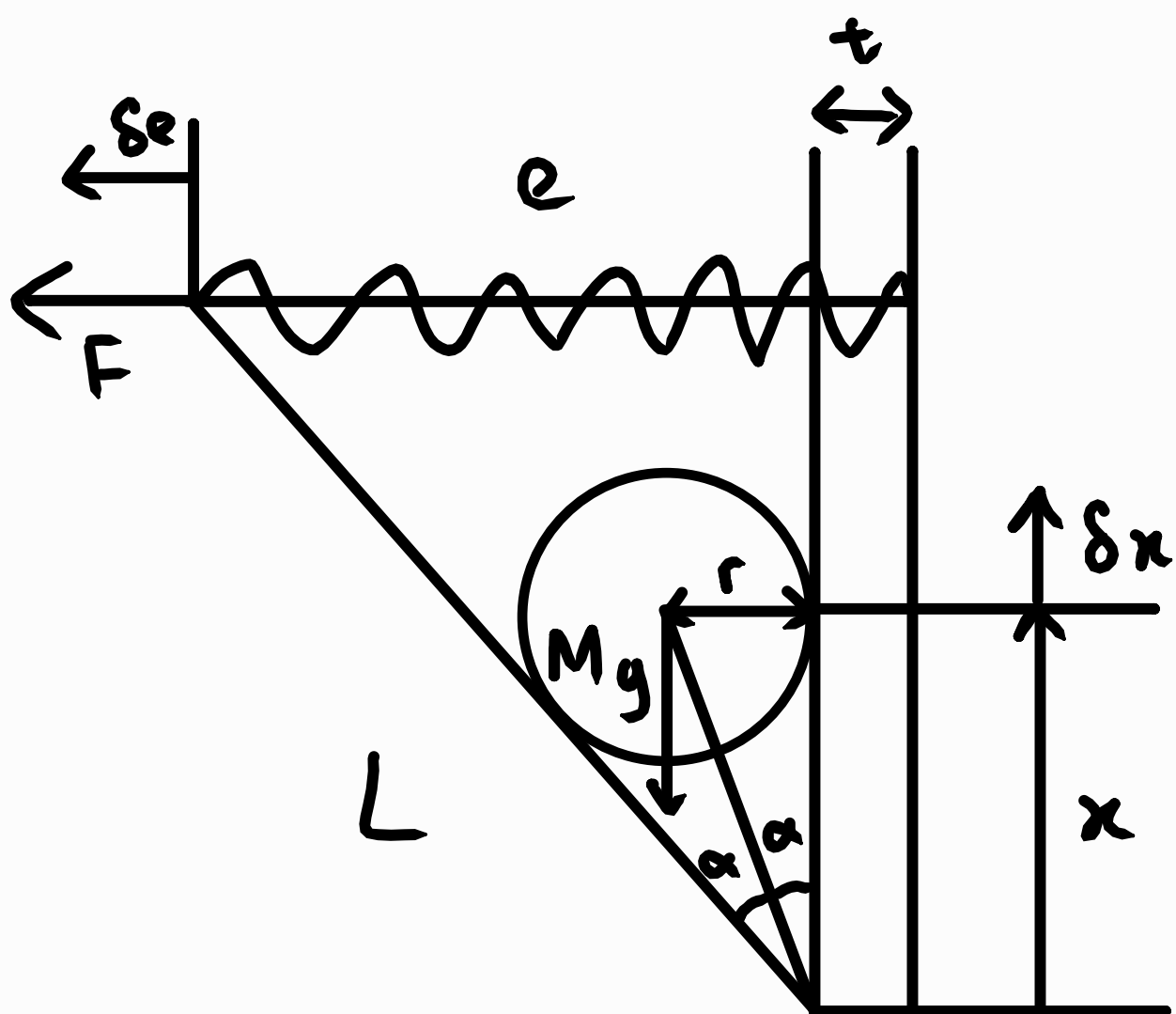
$$mg\ell \cos \theta - m\omega^2 \ell \sin \theta (b + \ell \cos \theta) = -k e a \cos \theta$$

$$m\omega^2 l \sin\theta (b + l \cos\theta) - mg l \cos\theta = k(2a)(1 - \sin\theta) a \cos\theta$$

$$w^2 m l \sin \theta (b + 2 \cos \theta) - mg l \cos \theta = 2a^2 k (1 - \sin \theta) \cos \theta$$

$$W = \frac{2a^2 k(1 - \sin \theta) \cos \theta + mg^2 \cos \theta}{m^2 \sin \theta (b + 2 \cos \theta)}$$

3)



$$\tan \alpha = \frac{r}{x}$$

$$x = \frac{r}{\tan \alpha}$$

$$e = L \sin 2\alpha + t - l_0$$

$$x = \frac{r}{\tan \alpha} = r \cot \alpha$$

$$\delta e = 2L \cos 2\alpha \delta \alpha$$

$$\delta x = -r \operatorname{cosec}^2 \alpha \delta \alpha$$

$$\delta W = \delta U$$

$$-Mg \delta x = F \delta e$$

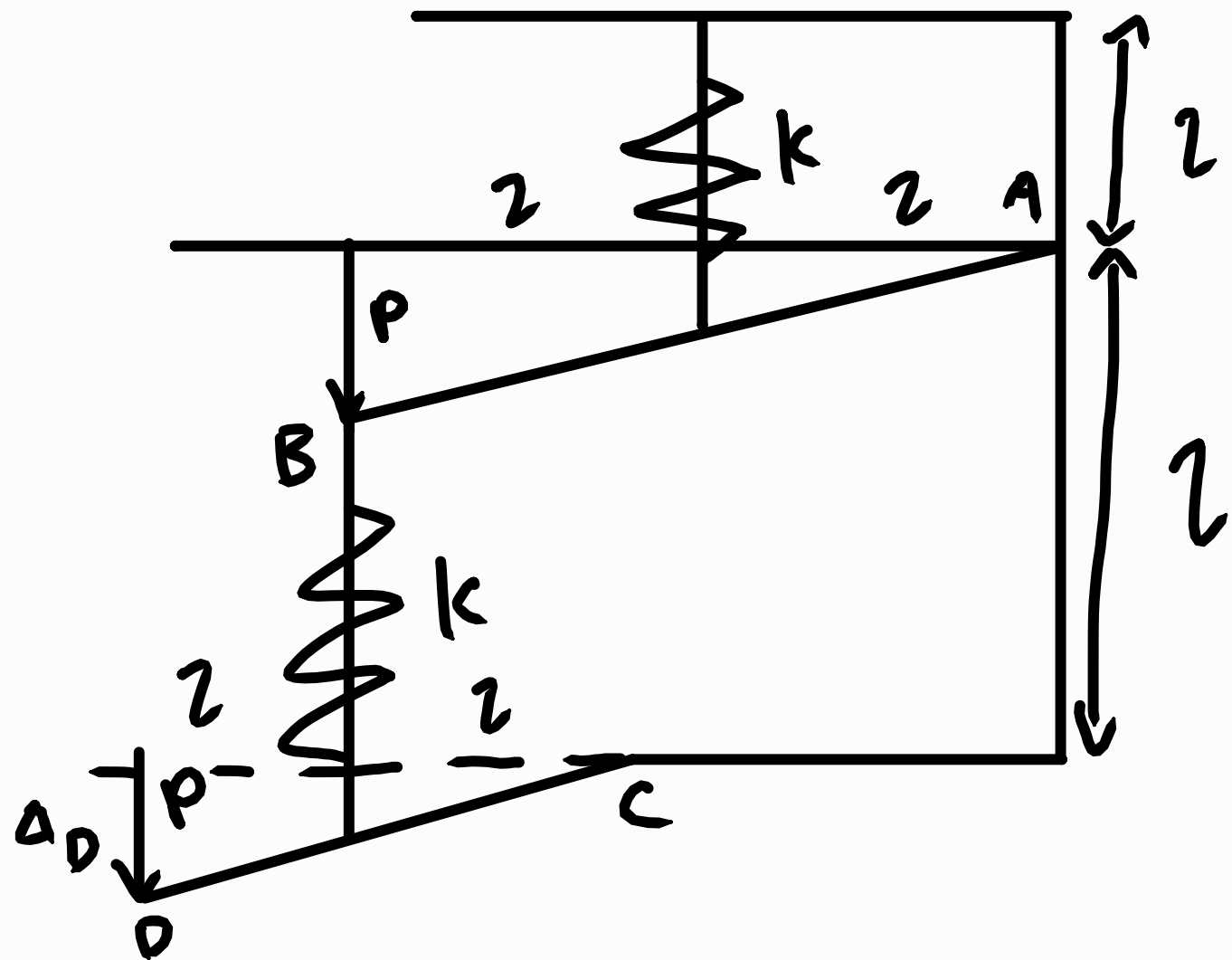
$$-Mg(-r \operatorname{cosec}^2 \alpha \delta \alpha) = F(2L \cos 2\alpha \delta \alpha)$$

$$\frac{Mgr}{\sin^2 \alpha} = 2K e L \cos 2\alpha$$

$$\frac{Mgr}{\sin^2 \alpha} = 2K (L \sin 2\alpha + t - l_0) L \cos 2\alpha$$

$$K = \frac{Mgr}{2L \cos 2\alpha \sin^2 \alpha (L \sin 2\alpha + t - l_0)}$$

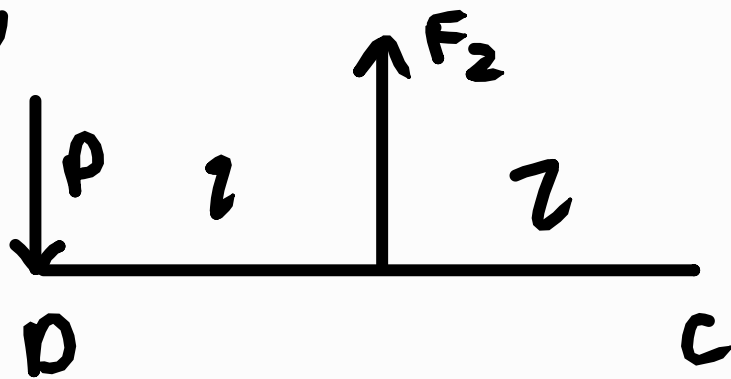
4)



$$\delta W^* = \delta U^*$$

$$\Delta_D \delta P_D = \frac{F_1}{k} \delta F_1 + \frac{F_2}{k} \delta F_2$$

On bar DC,

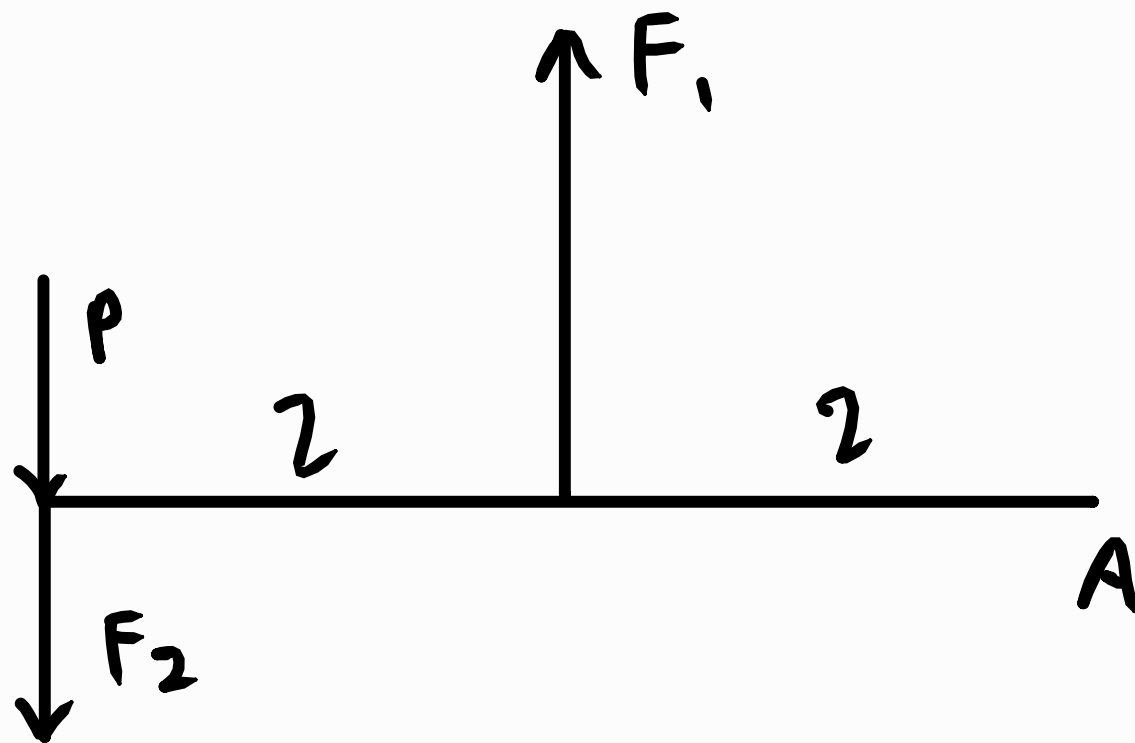


$$\sum M_C = 0$$

$$P(2) = F_2(1)$$

$$F_2 = 2P$$

4) On bar AB,

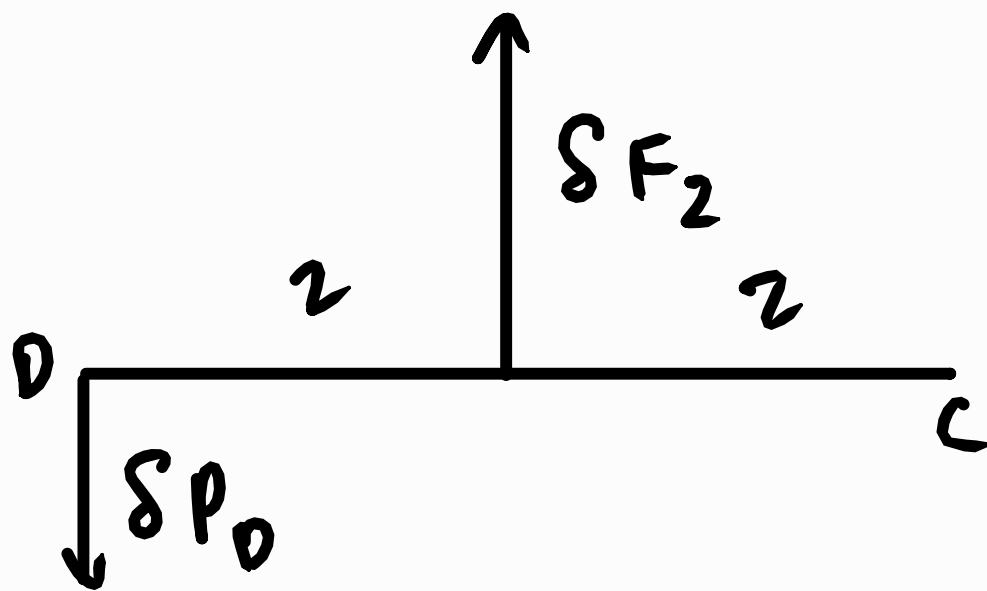


$$(P + F_2)(2x) = F_1 x$$

$$(P + 2P)(2) = F_1$$

$$F_1 = 6P$$

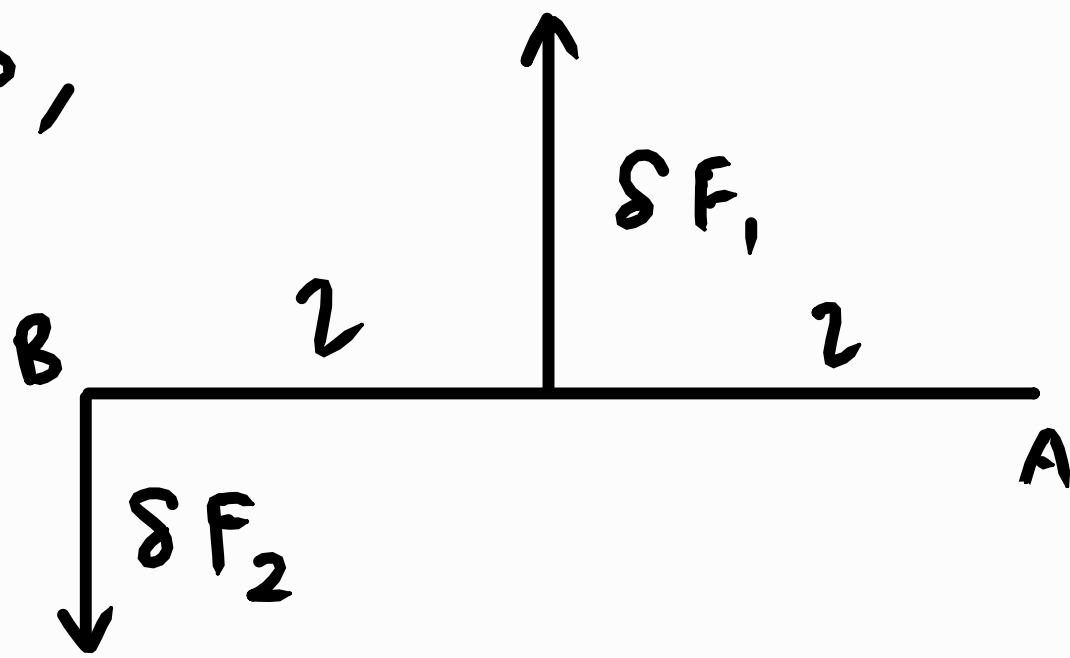
On bar CD,



$$\delta P_0 (2x) = \delta F_2 x$$

$$\delta F_2 = \delta P_0$$

4) On bar AB,



$$\delta F_2 (2l) = \delta F_1 (l)$$

$$2\delta F_2 = \delta F_1$$

$$\delta F_1 = 4\delta P_0$$

$$\therefore F_1 = 6P, F_2 = 2P, \delta F_1 = 4\delta P_0, \delta F_2 = 2\delta P_0$$

$$\Delta_D \delta P_0 = \frac{F_1}{k} \delta F_1 + \frac{F_2}{k} \delta F_2$$

$$\Delta_D \cancel{\delta P_0} = \frac{6P}{k} (4\cancel{\delta P_0}) + \frac{2P}{k} (2\cancel{\delta P_0})$$

$$\Delta_D = \frac{28P}{k}$$